Technical Report

Candidate Ranking with Word2Vec and Rule-Based Relevance

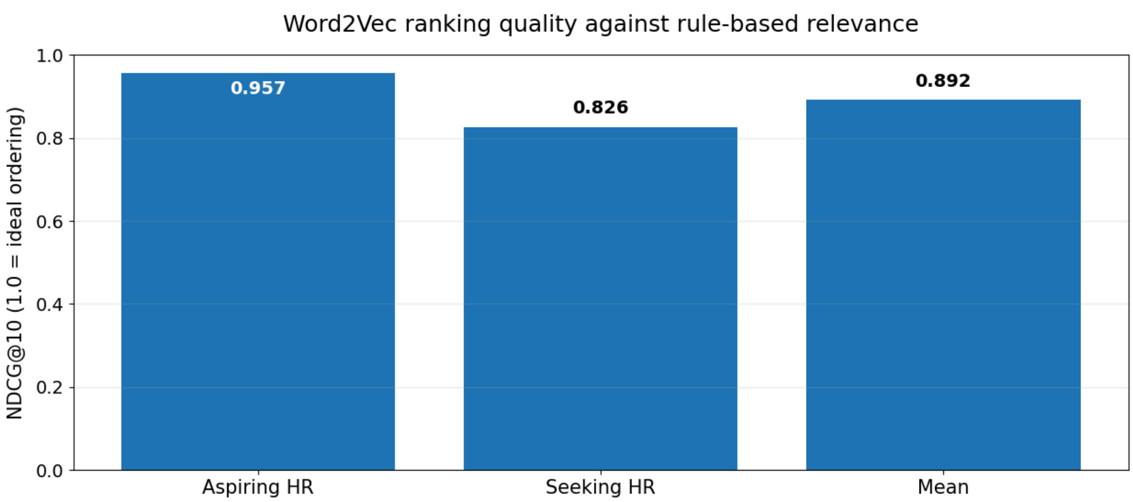
Executive Summary

This project evaluates how an already-sourced candidate pool can be ranked against the recruiter searches **"aspiring human resources"** and **"seeking human resources"**, then dynamically reranked when a recruiter stars an ideal candidate. Because ranking uses only `job_title`, evaluation is conducted on the **52 unique job titles** contained in the 104-record source dataset, preventing repeated titles from receiving extra weight while preserving the ability to map scores back to every candidate ID.

Two independent approaches are compared: pretrained **GoogleNews Word2Vec** semantic ranking and a transparent rule-based method, **R = H(0.70 + 0.30I)**, where H captures occupational relevance and I captures search-intent alignment. Word2Vec achieved NDCG@10 scores of **0.957** and **0.826** for the two searches, respectively. An independent 0-3 human relevance assessment showed **75.0% exact agreement** with the rule levels, **quadratic weighted kappa = 0.888**, and **Spearman correlation = 0.871**.

99.73%	0.892	30%
Word2Vec token coverage	Mean NDCG@10	Selected feedback influence

Figure 1. Main ranking result



NDCG@10 measures whether the titles judged most relevant by the rule-based benchmark appear near the top of the Word2Vec top ten. 1.0 represents the ideal ordering.

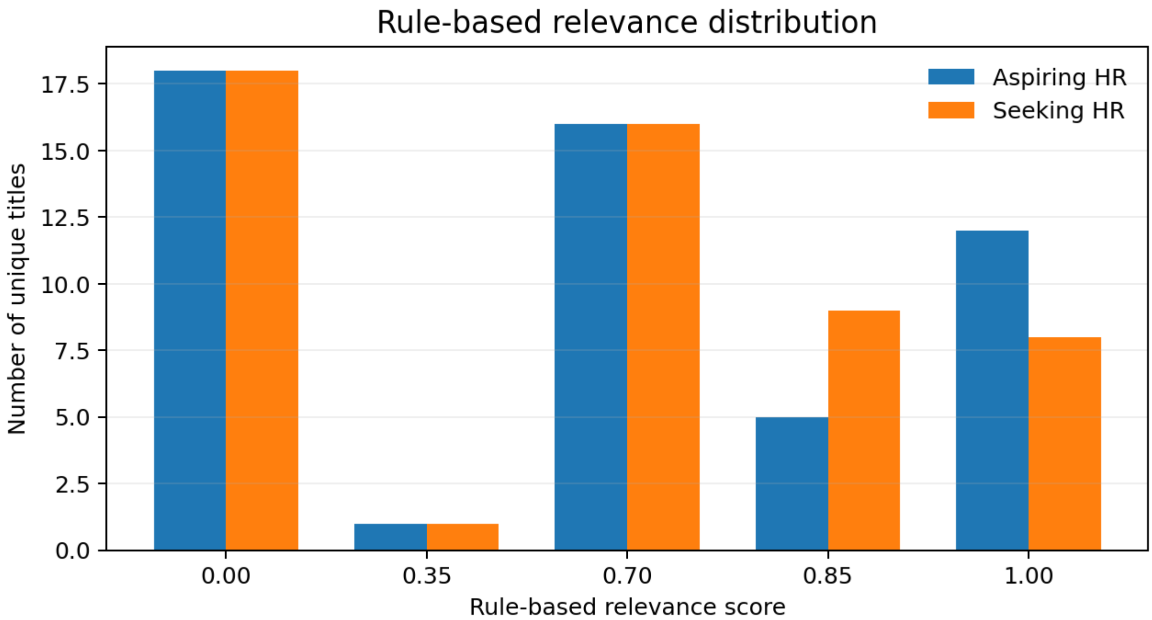
Dataset structure and design rationale

Exact-title repetition is material: 14 of 52 unique titles recur in the 104 rows, and the three most repeated titles each appear 7 times. After deduplication, 31/52 titles contain the phrase Human Resources and 34/52 contain direct HR evidence; 12 contain aspiring and 10 contain seeking. Common descriptors such as professional (8), manager (7), generalist (5), management (5), and specialist (4) coexist with sparse adjacent terms such as staffing (2) and recruiting, talent, payroll, benefits, and compensation (1 each). These frequencies support explicit rule signals while the varied, sparse vocabulary motivates Word2Vec; title-level evaluation prevents duplicates from dominating the metrics.

1. Ranking Behavior and Error Analysis

The rule-based method keeps occupational relevance separate from recruiter intent. Direct HR evidence receives the strongest role score, adjacent functions receive lower relevance, and titles with no defensible HR evidence receive zero. Intent then distinguishes whether a title explicitly reflects *aspiring* or *seeking*. The multiplicative structure prevents generic intent language from manufacturing role relevance, while a directionality rule prevents employer advertisements from being treated as candidates.

Figure 2. Rule-based relevance distribution



The underlying HR relevance structure remains stable; changing the active query mainly redistributes the strongest 0.85 and 1.00 scores according to intent.

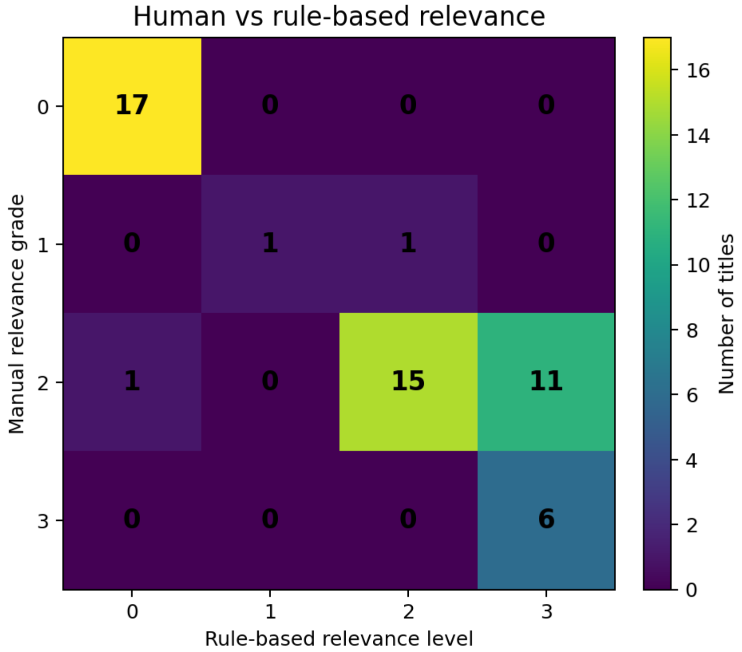
Observed Word2Vec failure cases

Example	Why Word2Vec ranked it highly	Why rule logic disagreed
ID 75 Employer advertisement	Contains “seeking Human Resources”	Employer seeking candidates, not a candidate seeking HR
ID 92 Customer Service / Patient Care	Generic “seeking” language raises semantic similarity	No defensible HR occupational relevance
Long HR titles	Relevant HR terms are averaged with unrelated words	Explicit role evidence is not diluted by title length

These disagreements are useful because they expose limitations in semantic similarity rather than hiding them: Word2Vec captures related meaning well, but it does not reliably infer **who is seeking whom** and can dilute strong terms in long titles.

2. Independent Human Validation

The 52 unique titles were independently graded from 0 to 3 for HR relevance. The labels were kept separate from both ranking methods and were used only to check whether the rule structure broadly aligned with human judgement.



75.0%

Exact agreement

0.888

Quadratic weighted kappa

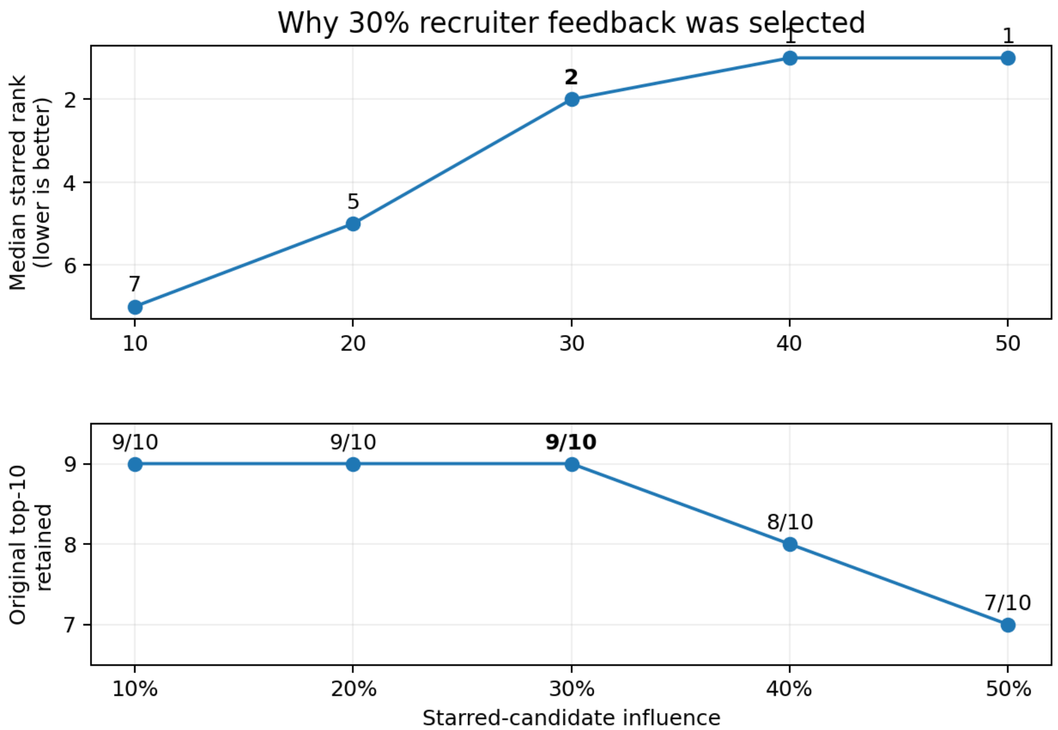
0.871

Spearman correlation

Most disagreement occurs between neighbouring relevance levels; all manual grade-0 titles and all manual grade-3 titles align with the corresponding rule extremes.

3. Recruiter Feedback: Why 30%?

The feedback stage shifts the original Word2Vec query toward the starred candidate, then reranks all 52 unique titles. The weight was tested systematically across **34 plausible ideal-title/query scenarios**, not selected from a single demonstration.



At 30% influence, the median starred title moves to rank 2 while 9 of the original top 10 remain. Higher weights move the star more aggressively but increasingly disturb the shortlist.

4. What Recruiter Feedback Changes

The seeking-HR demonstration shows the business meaning of the feedback mechanism. Starring ID 99 moves the ideal candidate from rank 8 to rank 1, while a second relevant HR title rises and an unrelated non-HR title falls. NDCG@10 improves from **0.826 to 0.904**.

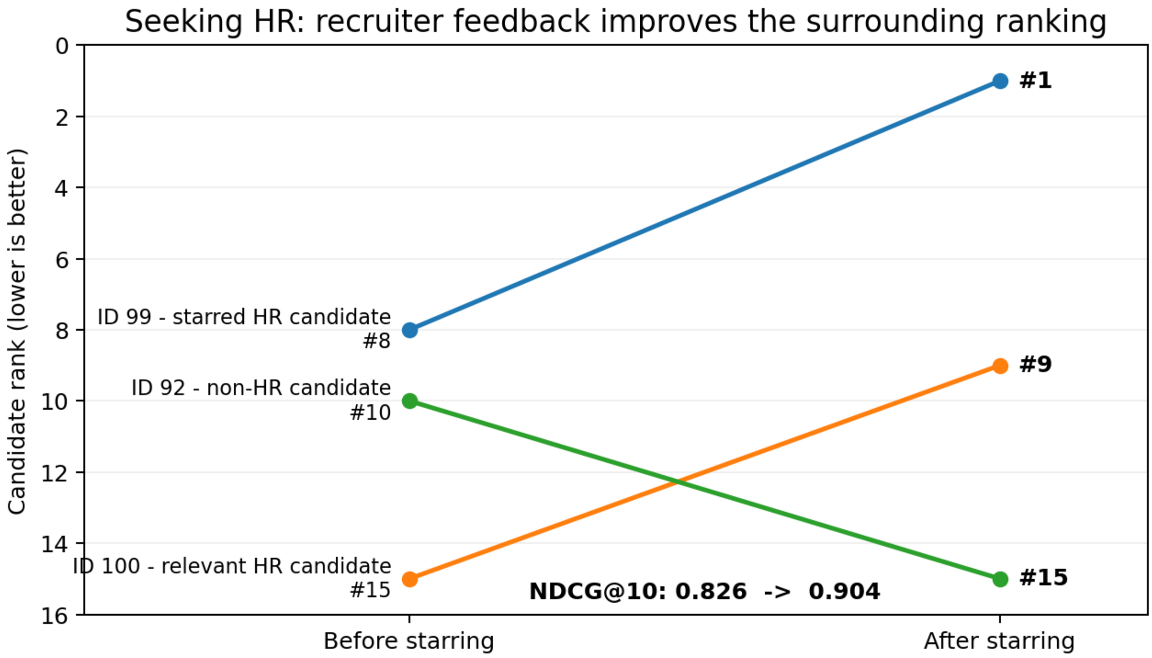


Figure 5. Seeking-HR before/after movement at the selected 30% feedback weight.

Practical interpretation

Use rules for	Use Word2Vec for	Use recruiter feedback for
Flagging profiles with no defensible role relevance; handling obvious directionality errors.	Relative semantic ranking among plausible candidates; flexible response to changing recruiter queries.	Personalizing the shortlist toward a demonstrated ideal without retraining a model.

A universal cosine-similarity cutoff is not recommended because absolute similarity values can change across roles and queries. The safer design is to use explicit role relevance to flag obvious non-matches, then use Word2Vec for relative ordering. The same rule structure can scale beyond HR by redefining occupational relevance and search intent for each role.

Limitations. Mean-pooled Word2Vec does not reliably understand grammatical directionality, long titles can dilute important words, and job-title-only ranking cannot use skills or experience that are not expressed in the title. Pretrained embeddings and recruiter feedback can also introduce bias, so human oversight remains necessary.

Conclusion

Overall, the project shows that **Word2Vec can rank candidates effectively, while rule-based relevance improves interpretability and catches obvious semantic errors**. Combined with recruiter-driven reranking, the approach can reduce manual screening effort, surface stronger candidates earlier, and keep human judgement in control.